

# Trial Examination

VCE Algorithmics (HESS) · 60 marks · 120 minutes · timed conditions

Name: \_\_\_\_\_

## Question 1 — Data modelling and ADTs (8 marks)

A hospital tracks patient transfers between wards. Transfers happen in order; staff need to (i) look up a patient's current ward instantly, (ii) replay the full transfer history in order, (iii) find whether two wards are directly connected by a corridor.

a.) Select an ADT (or combination) for each requirement, with signatures or key operations, and justify each choice with its operation cost. (6)

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b.) The hospital adds: “find the fewest-corridor route between any two wards.” Name the algorithm and its complexity on adjacency lists. (2)

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## Question 2 — Pseudocode and complexity (12 marks)

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1 Algorithm Dedupe(L):  
2   result ← empty list  
3   Foreach x in L Do  
4     found ← False  
5     Foreach y in result Do  
6       If x = y Do  
7         found ← True  
8     If Not found Do  
9       append x to result  
10  Return result
```

a.) Worst-case complexity with justification. (3)

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b.) Describe the best-case input and its complexity. (2)

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c.) Rewrite Dedupe in VCAA pseudocode using a dictionary to achieve  $O(n)$ , and justify the new bound. (4)

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d.) Using the formal definition of Big-O, show the original algorithm's cost  $f(n) = n^2 + 4n$  is  $O(n^2)$ . (3)

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### Question 3 — Recursion and Master Theorem

(8 marks)

a.) An algorithm splits a problem of size  $n$  into 3 subproblems of size  $n/3$  and combines in  $O(n)$ . Write the recurrence and solve it, naming the case. (3)

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b.) Solve  $T(n) = 2T(n-1) + 1$ ,  $T(1) = 1$ , by expansion, showing the pattern. (3)

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c.) State why the Master Theorem does not apply in (b). (2)

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## Question 4 — Graph algorithms with correctness

(8 marks)

a.) Dijkstra's algorithm fails on graphs with negative edge weights. Explain why, referring to the greedy finalisation step. (3)

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b.) Prim's algorithm: state the invariant it maintains, and the exchange-argument idea behind its correctness. (3)

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c.) State PageRank's per-iteration and total complexity, defining any symbols. (2)

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## Question 5 — Intractability and heuristics

(12 marks)

A festival must assign 400 performers to stages over a weekend, minimising clashes for attendees who ranked their favourites.

a.) Argue this resembles an NP-hard problem, naming a studied problem it relates to. (2)

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b.) Explain why the organisers should not wait for an exact optimum, quantitatively. (2)

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c.) Design a simulated annealing approach: representation, neighbourhood, objective, and cooling behaviour. (4)

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d.) After deployment, attendees ask whether the schedule is optimal. Write the two-sentence honest answer, and name one form of evidence you could offer. (4)

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## Question 6 — Computability and AI (12 marks)

a.) Explain the significance of the Entscheidungsproblem to the birth of computer science. (2)

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b.) Prove the Halting Problem is undecidable. (4)

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c.) Describe the Chinese Room and evaluate the robot reply in 3–4 sentences. (4)

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d.) A network with step activations: inputs  $x_1 = 0, x_2 = 1$ ;  $h_1$  (weights 0.9,  $-0.2$ , bias 0.0),  $h_2$  (weights 0.3, 0.6, bias  $-0.5$ ); output (weights 0.5, 1.0, bias  $-0.9$ ). Evaluate with working. (2)

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*End of trial examination.*